Scalability & Future Plan

Date	01 July 2026
Team ID	xxxxx
Project Name	Rising Waters: A Machine Learning Approach to Flood Prediction
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	No real-time IoT / weather API integration	System relies on manually entered or historical data, delaying live flood monitoring	High
2	No authentication or session layer	Predictions are accessible to anyone using the app, limiting accountability and access control	Medium
3	Single-instance Flask deployment (no load balancing)	Limits the number of concurrent users the system can reliably serve	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Single Flask/Gunicorn instance suitable for a small pilot group	Deploy on IBM Cloud with auto-scaling and load balancing to support concurrent disaster-response teams
2	Data Storage	Local CSV files and serialized model/scaler (floods.save) stored on disk	Migrate to managed cloud storage (IBM Cloud Object Storage / Db2) for historical data and model versioning
3	Performance	Model is loaded and run per request without caching	Introduce model caching and batch-prediction endpoints to reduce latency under higher load
4	Security	No authentication or encryption implemented in current scope	Add role-based authentication and HTTPS/TLS encryption ahead of production deployment

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Integrate real-time meteorological / weather API	Q3 2026	Enables live, automated flood risk monitoring instead of manual data entry
Phase 2	Add authentication & role-based access control	Q4 2026	Restricts access to authorized disaster management personnel
Phase 3	Deploy scalable architecture on IBM Cloud with auto-scaling	Q1 2027	Supports higher concurrent usage across multiple regions
Phase 4	Add SMS / mobile push notifications for evacuation alerts	Q2 2027	Faster, wider reach for early warnings to at-risk communities